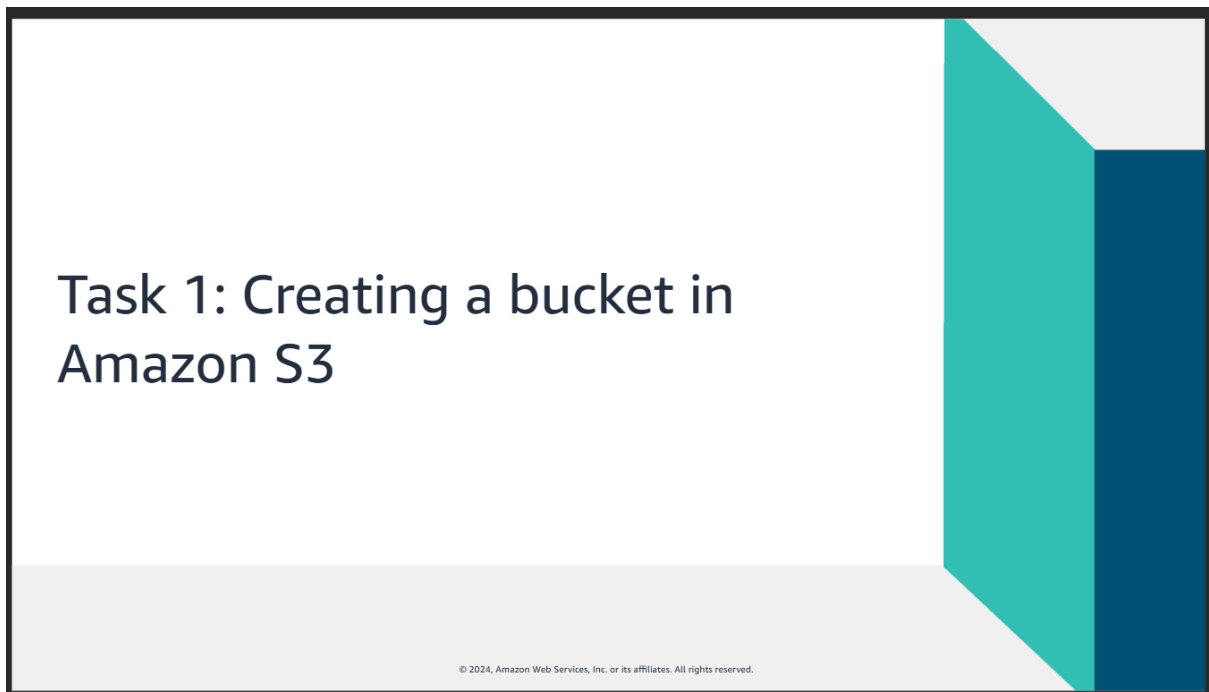
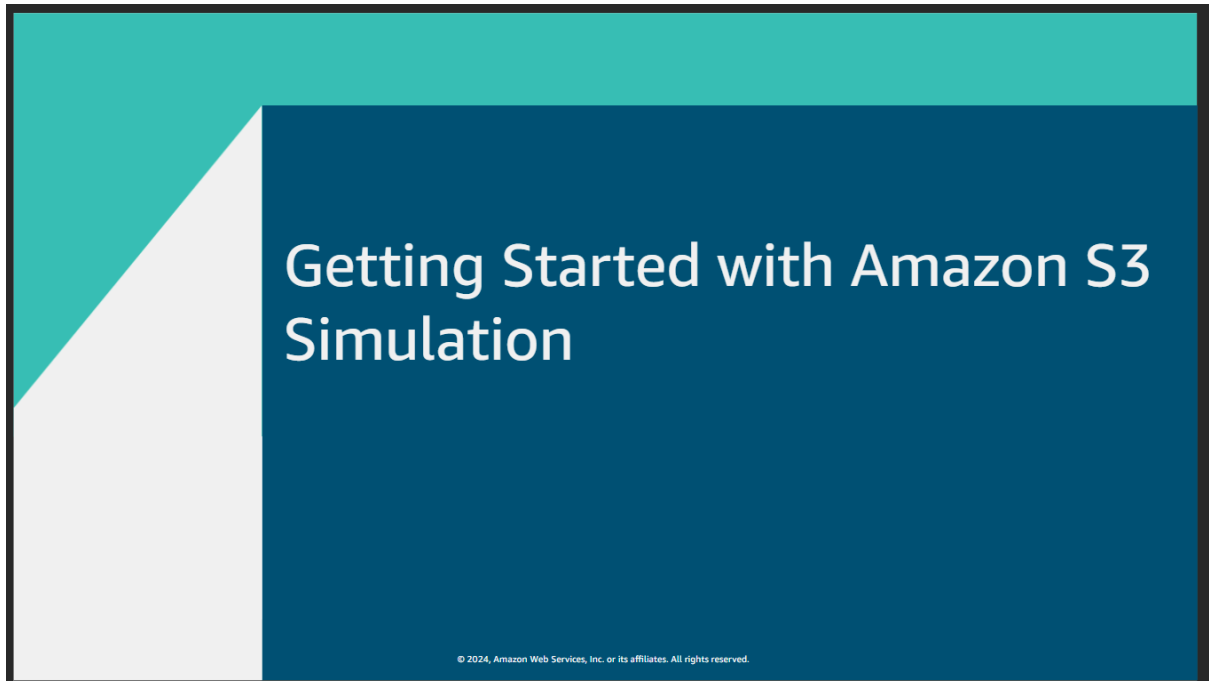
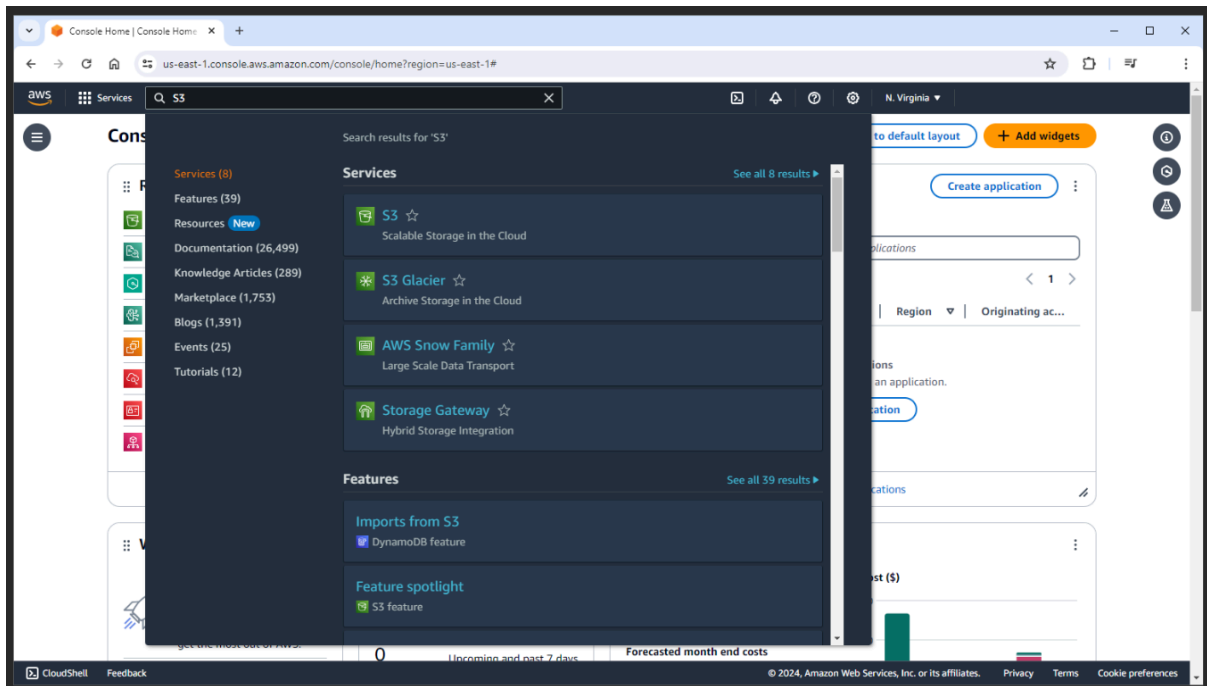


CC Assignment 1. Part-2

Title: AWS Educate - Getting started with Storage (S3)

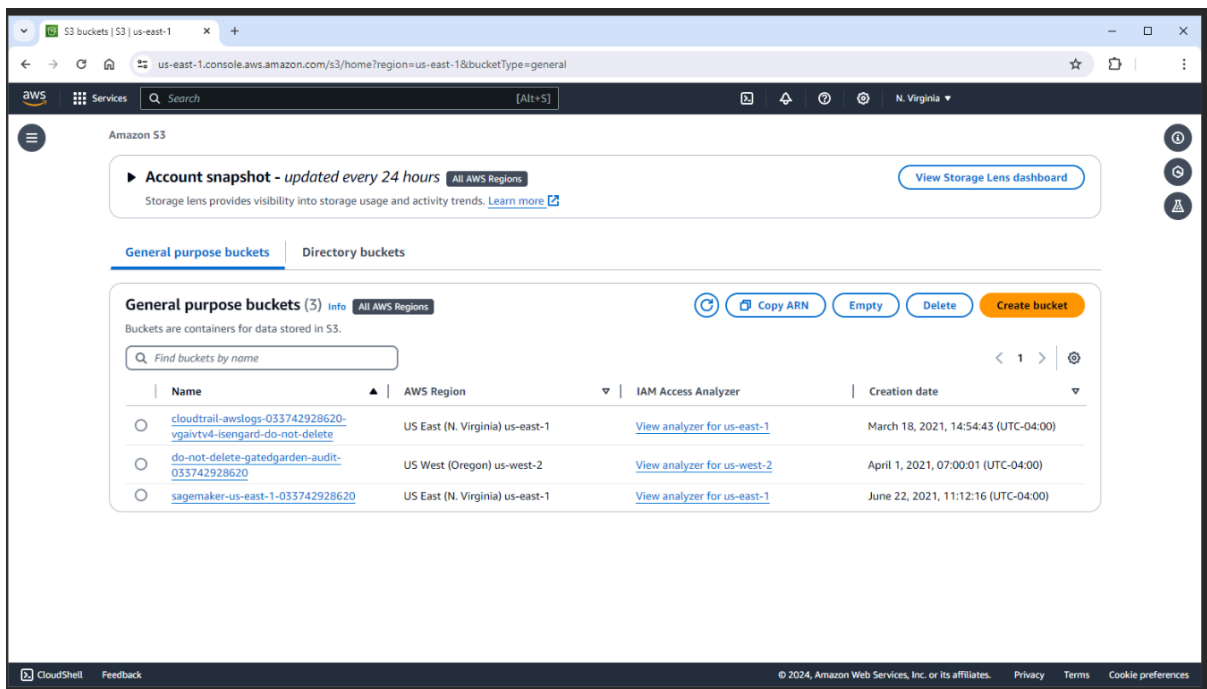
1. In the AWS Management Console, choose the search bar and enter S3.





2. Then choose S3 from the search results.

3. Choose Create bucket.



4. For bucket name, enter sim-website.

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory - New**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

myawsbucket

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

5. Choose the scroll bar to scroll down to Object Ownership.
6. For Object Ownership, choose ACLs enabled. Keep the default Bucket owner preferred selected.

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☐ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

Object Ownership

☒ **Bucket owner preferred**
If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.

☐ **Object writer**
The object writer remains the object owner.

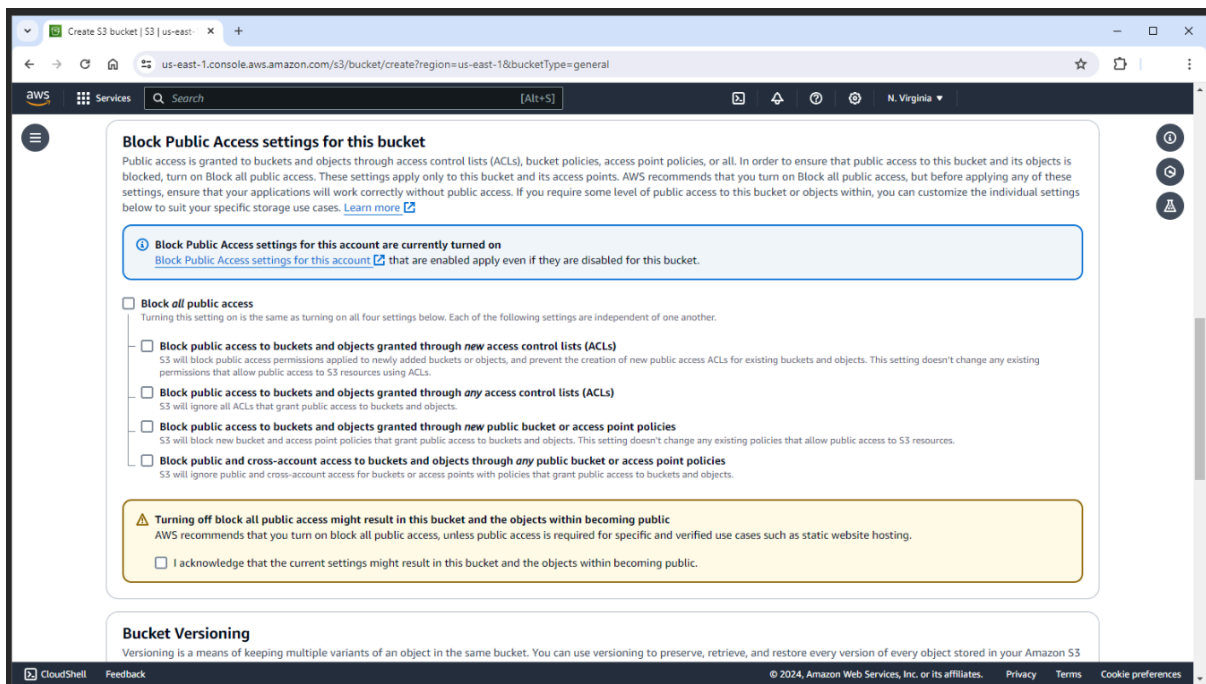
If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is required for object uploads. [Learn more](#)

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

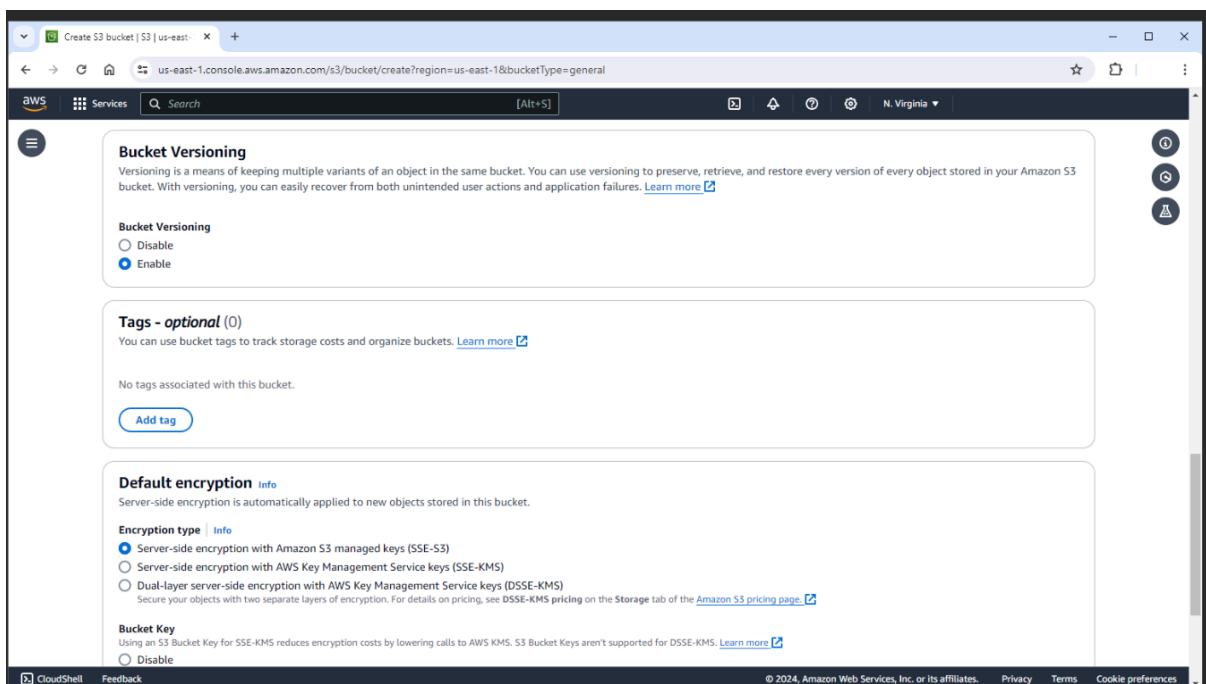
Block Public Access settings for this account are currently turned on
[Block Public Access settings for this account](#) that are enabled apply even if they are disabled for this bucket.

7. Choose the scroll bar to scroll down to Block Public Access settings for this bucket.



9. Choose the scroll bar to scroll down to Bucket Versioning.

10. For bucket versioning, choose "Enable."



11. For tags, choose "Add tag" and enter the following:

o Key: Department

o Value: Marketing

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Disable

☒ Enable

Tags - optional (1)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

Key

Department

Value - optional

Marketing

[Remove](#)

[Add tag](#)

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the [Storage](#) tab of the [Amazon S3 pricing page](#).

Bucket Key

12. Choose the scroll bar to scroll down.

13. Choose "Create bucket."

Successfully created bucket "sim-website"

To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Account snapshot - updated every 24 hours [All AWS Regions](#) [View Storage Lens dashboard](#)

Storage lens provides visibility into storage usage and activity trends. [Learn more](#)

General purpose buckets | **Directory buckets**

General purpose buckets (4) [Info](#) [All AWS Regions](#)

[Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

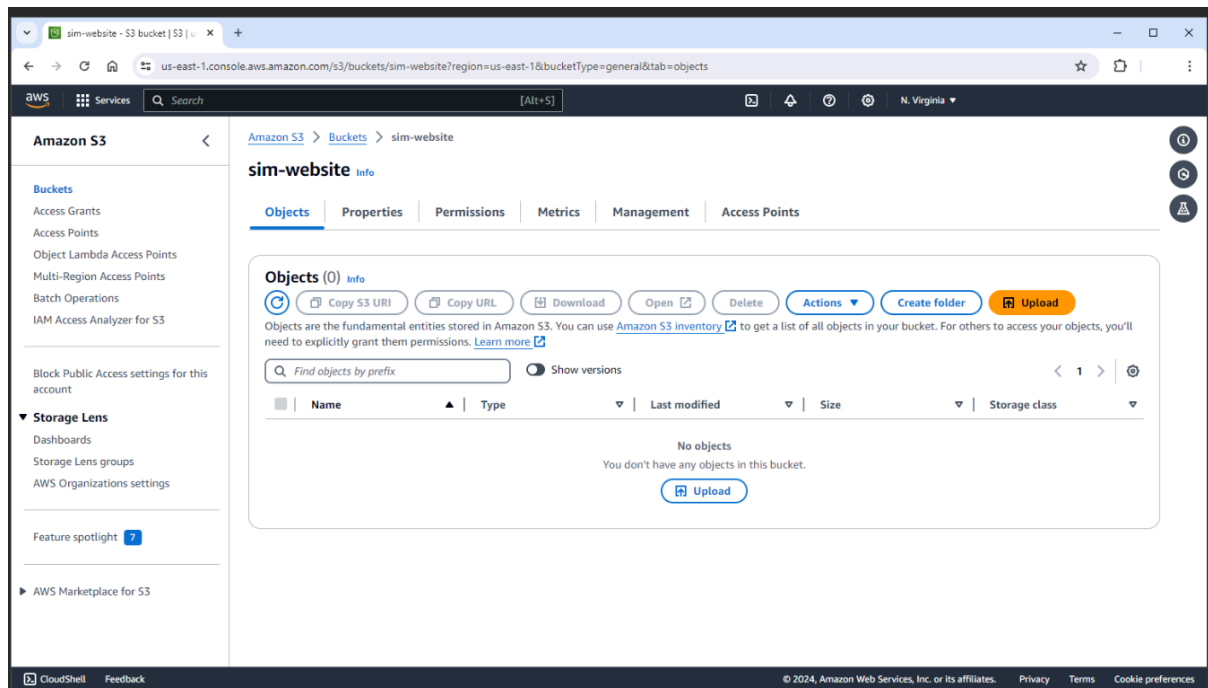
	Name	AWS Region	IAM Access Analyzer	Creation date
☐	cloudtrail-awslogs-033742928620-vgaivt4-isengard-do-not-delete	US East (N. Virginia) us-east-1	View analyzer for us-east-1	March 18, 2021, 14:54:43 (UTC-04:00)
☐	do-not-delete-gatedgarden-audit-033742928620	US West (Oregon) us-west-2	View analyzer for us-west-2	April 1, 2021, 07:00:01 (UTC-04:00)
☐	sagemaker-us-east-1-033742928620	US East (N. Virginia) us-east-1	View analyzer for us-east-1	June 22, 2021, 11:12:16 (UTC-04:00)
☐	sim-website	US East (N. Virginia) us-east-1	View analyzer for us-east-1	June 4, 2024, 15:53:27 (UTC-04:00)

This completes Task 1: Creating a bucket in Amazon S3. Choose Continue.

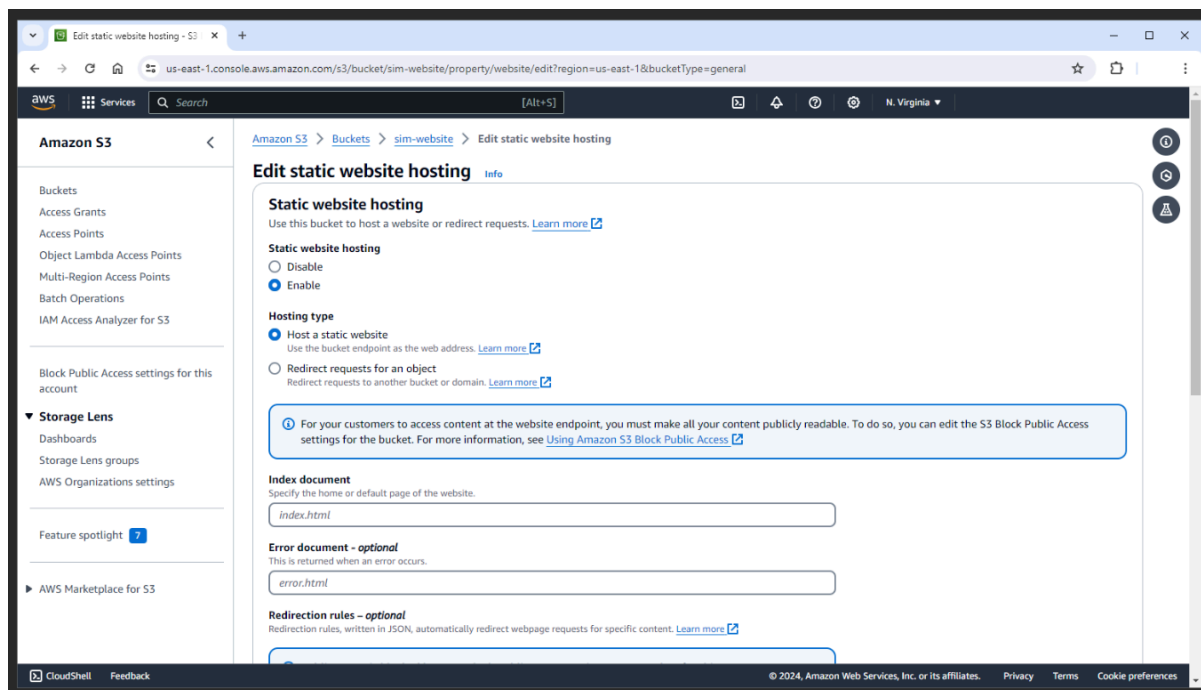
Task 2: Configuring a static website on Amazon S3

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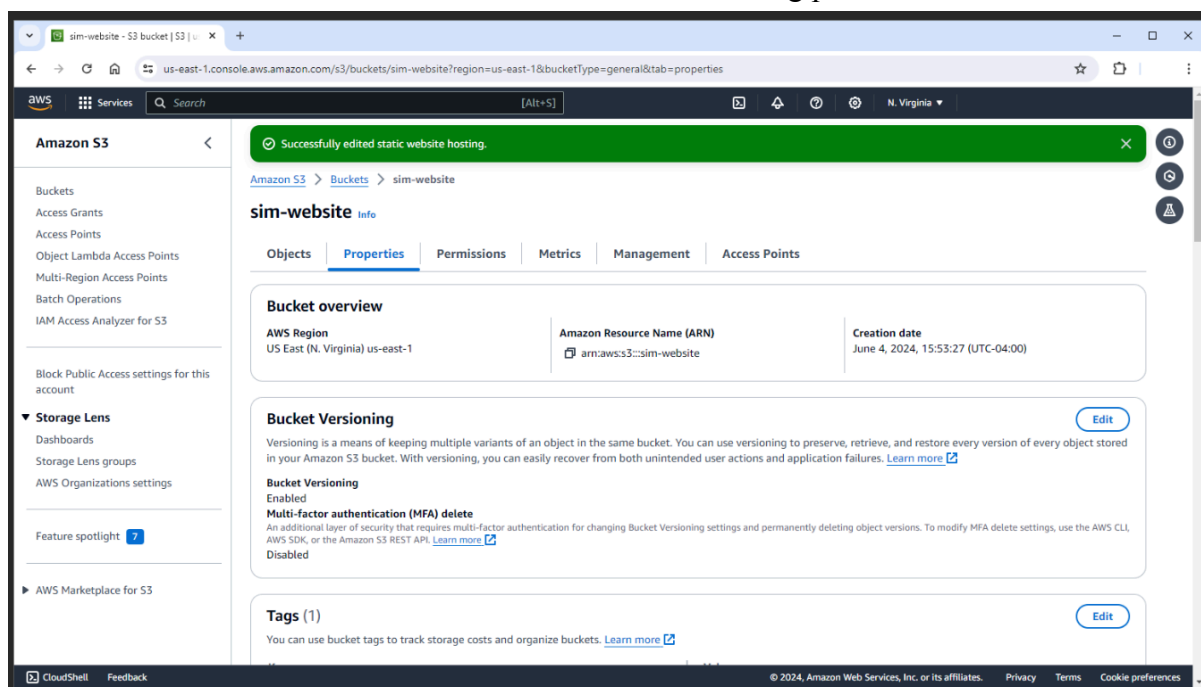
14. In the list of your buckets, choose the name of the bucket that you just created, sim-website.



15. Choose the Properties tab.
16. Choose the scroll bar to scroll to the static website hosting panel.
17. Choose "Edit" in the static website hosting panel.
18. Choose "Enable."
19. For hosting type, keep the default setting, host a static website.
20. Configure the following settings:
 - o Index document: Enter index.html
 - o Error document: Enter error.html

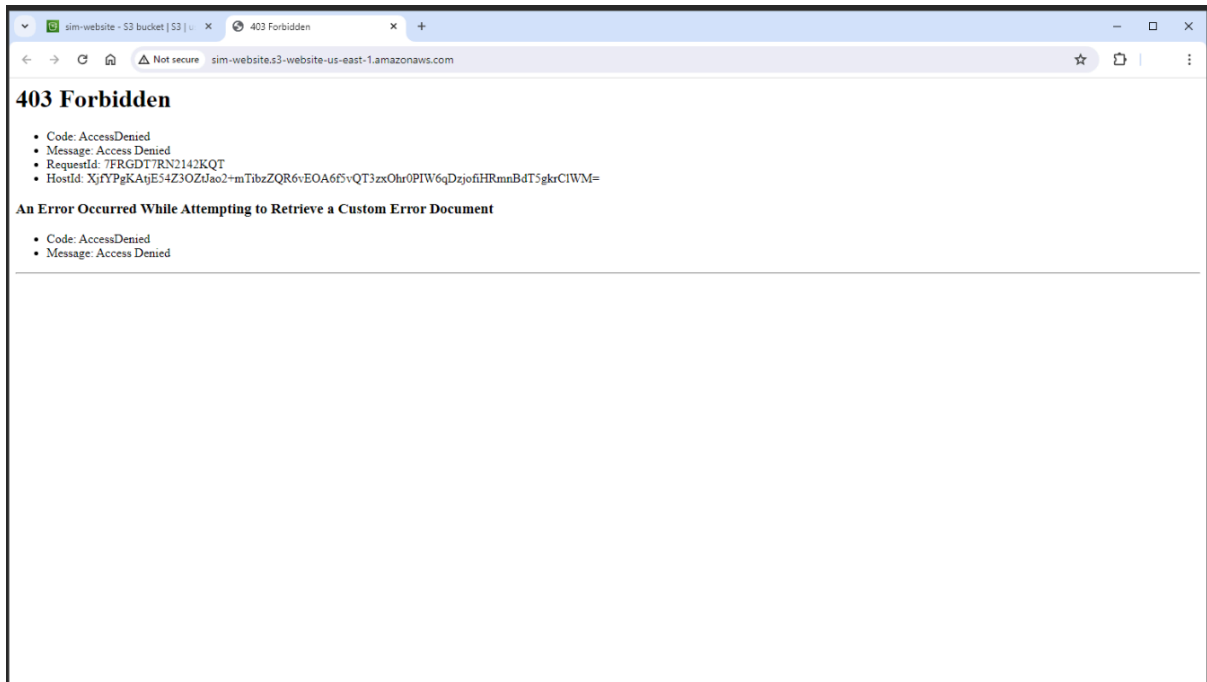


21. Choose the scroll bar to scroll down.
22. Choose Save changes.
23. Choose the scroll bar to scroll to the Static website hosting panel.

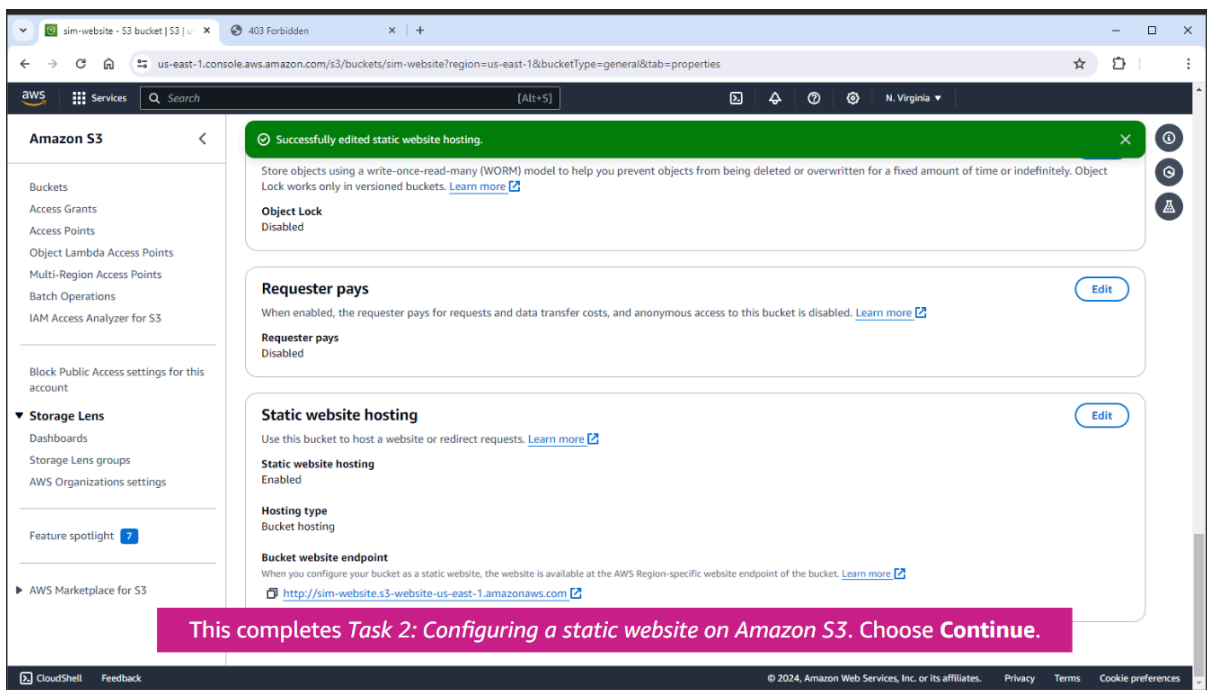


24. In the static website hosting panel under "Bucket website endpoint, choose the link.

A new tab opens where you receive a 403 Forbidden message because you have not yet configured the bucket permissions. You can return to it later.



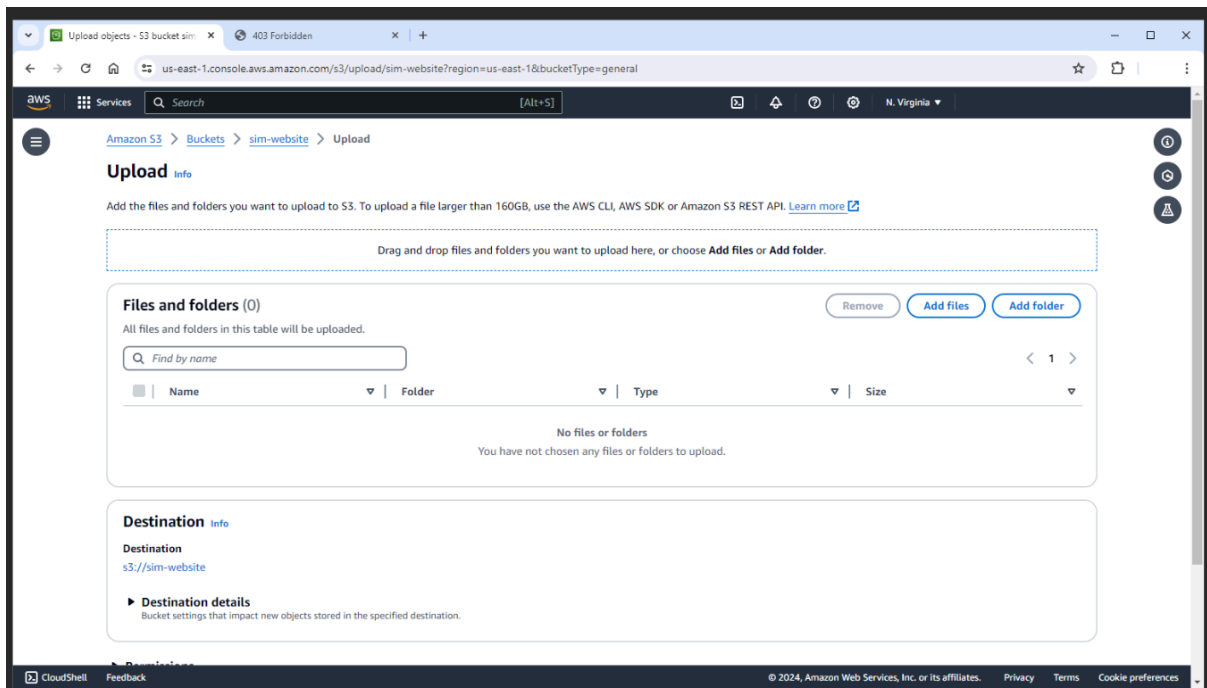
25. Choose the AWS Management Console tab on your browser.
You have configured your bucket to host a static website.



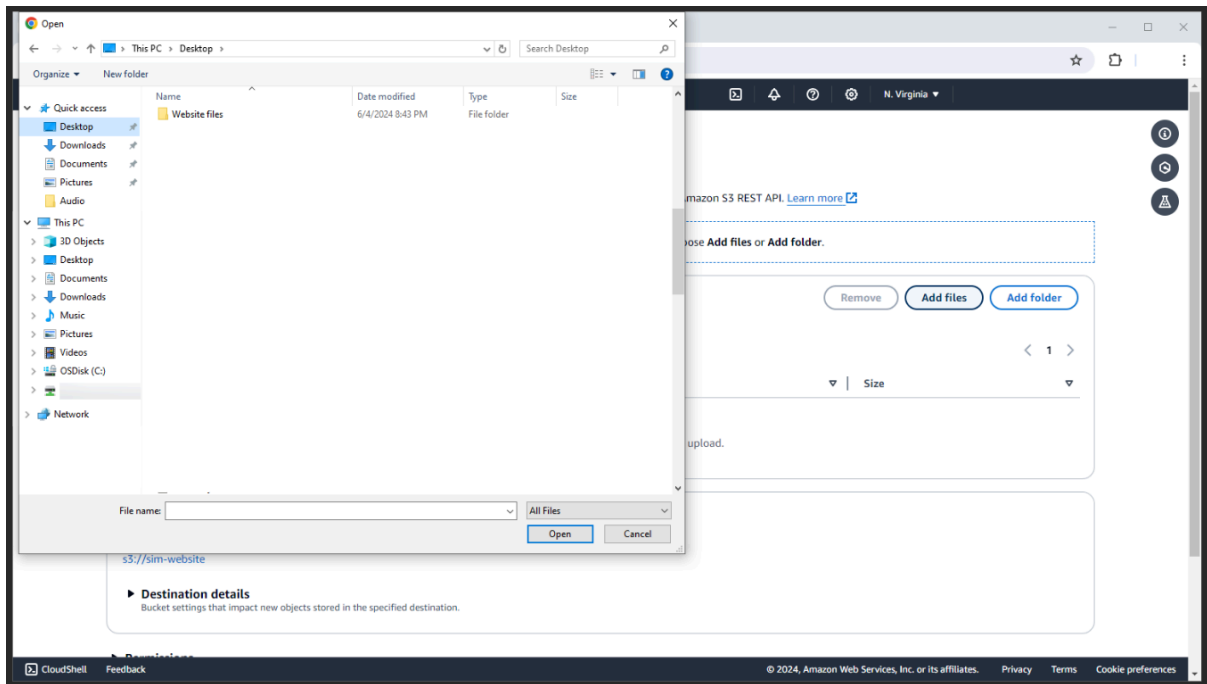
Task 3: Uploading content to your bucket

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26. Choose the scroll bar to scroll to the top of the page, and choose the Objects tab.
27. Choose Upload.

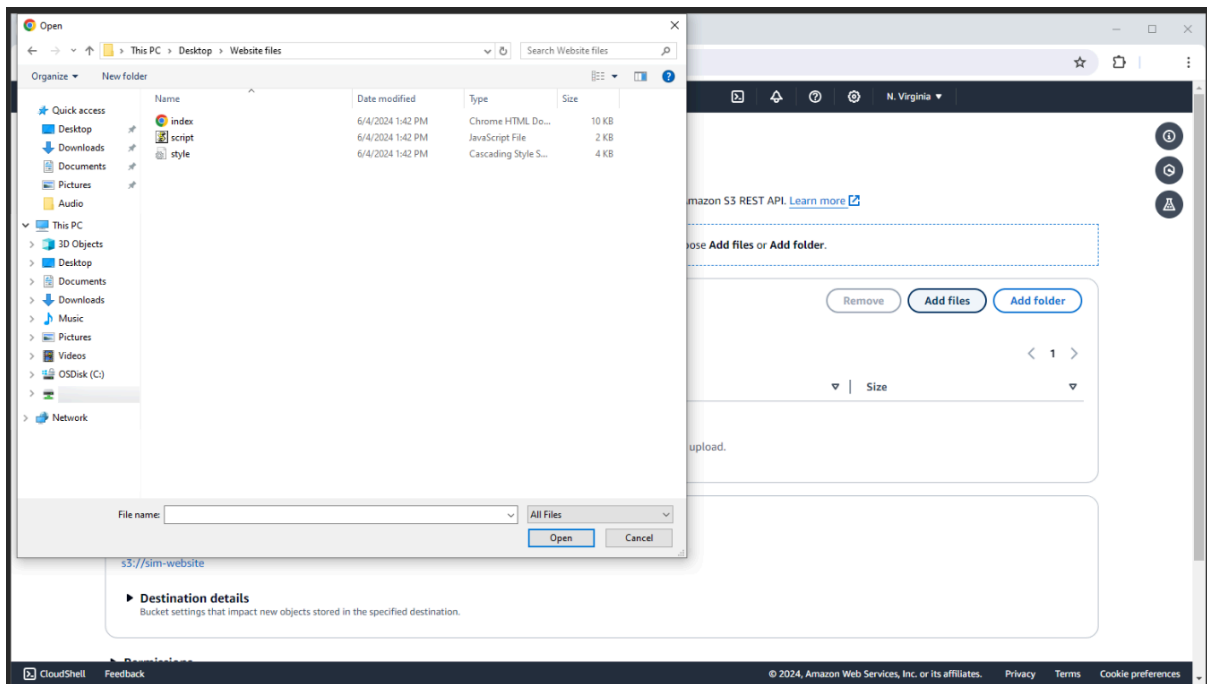


28. Choose Add files.
29. Choose the Website files folder, and choose Open to open the folder.

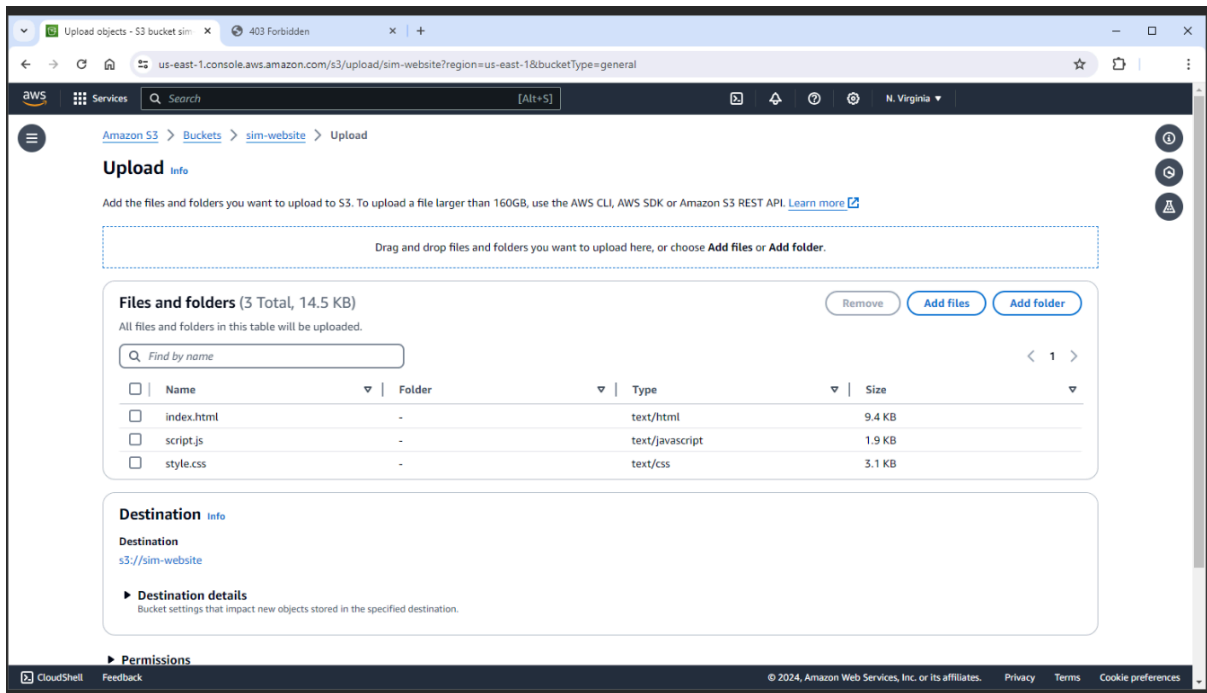


30. Use your mouse to choose each of the following files: index.html, script.js, and style.css (order does not matter). Then choose Open.

31. Choose the scroll bar to scroll down.

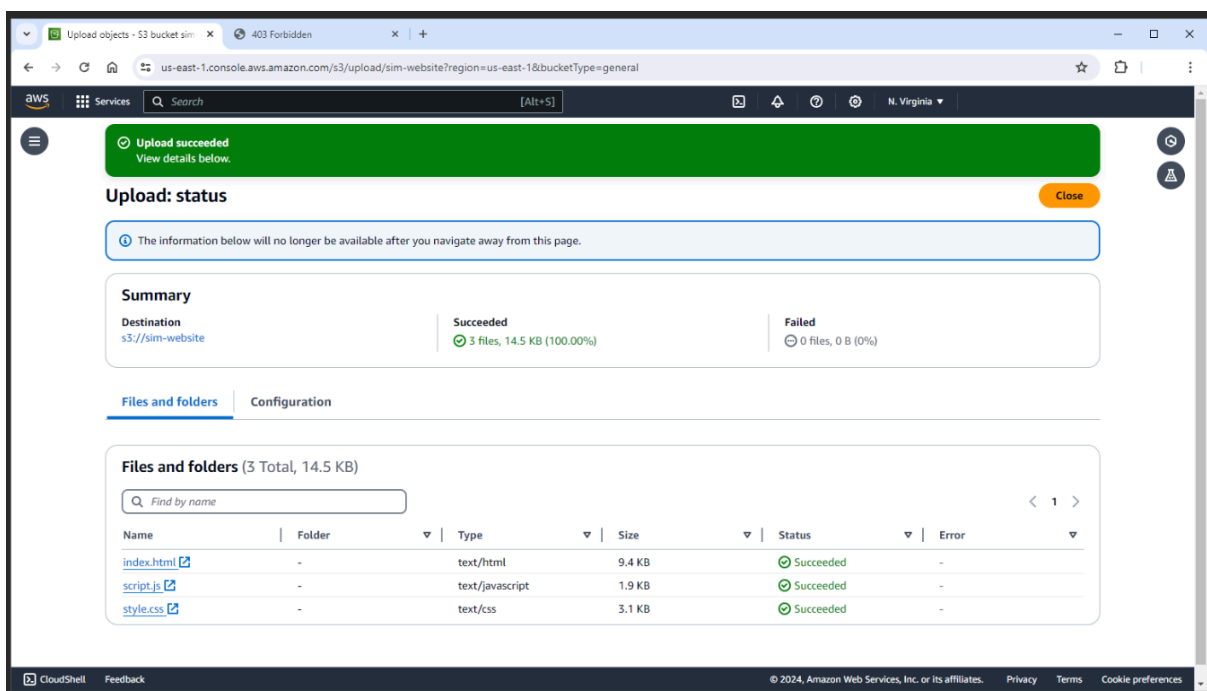


32. Choose Upload.



33. Choose Close.

You have successfully uploaded your website file to your sim-website bucket.



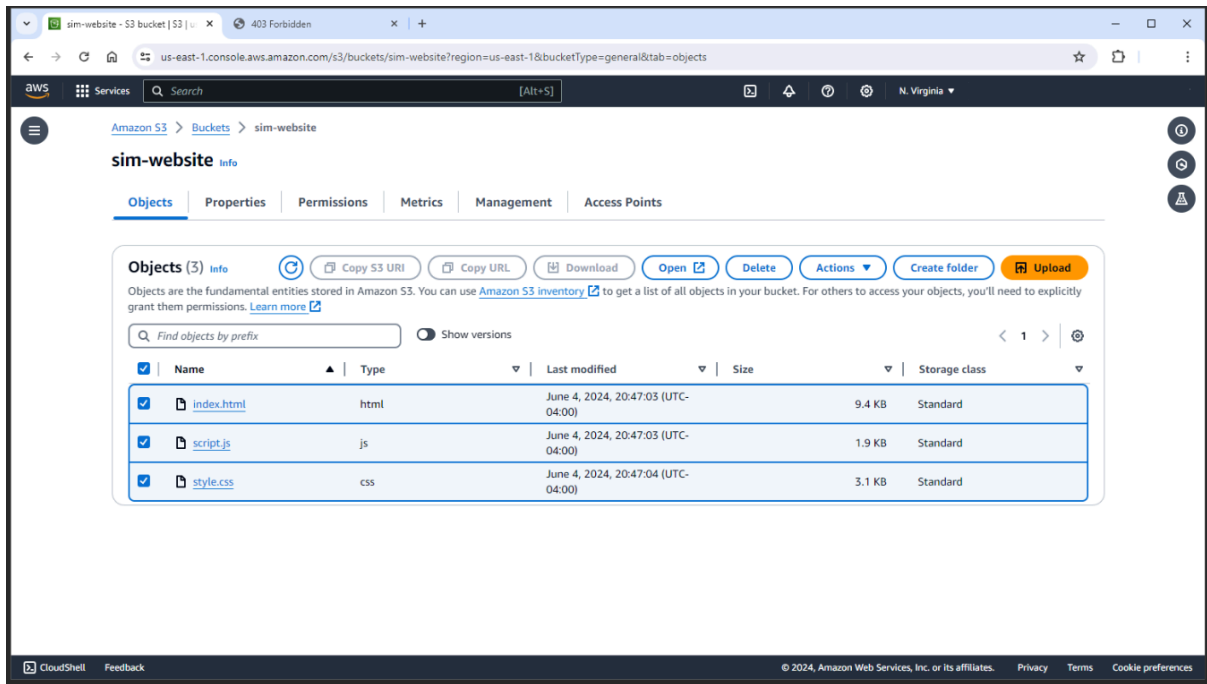
The screenshot shows the Amazon S3 console interface for a bucket named 'sim-website'. The 'Objects' tab is selected, displaying a list of three objects: 'index.html' (9.4 KB), 'script.js' (1.9 KB), and 'style.css' (3.1 KB). All objects are of 'Standard' storage class and were last modified on June 4, 2024. A pink banner at the bottom of the console area states: 'This completes Task 3: Uploading content to your bucket. Choose Continue.'

☐	Name	Type	Last modified	Size	Storage class
☐	index.html	html	June 4, 2024, 20:47:03 (UTC-04:00)	9.4 KB	Standard
☐	script.js	js	June 4, 2024, 20:47:03 (UTC-04:00)	1.9 KB	Standard
☐	style.css	css	June 4, 2024, 20:47:04 (UTC-04:00)	3.1 KB	Standard

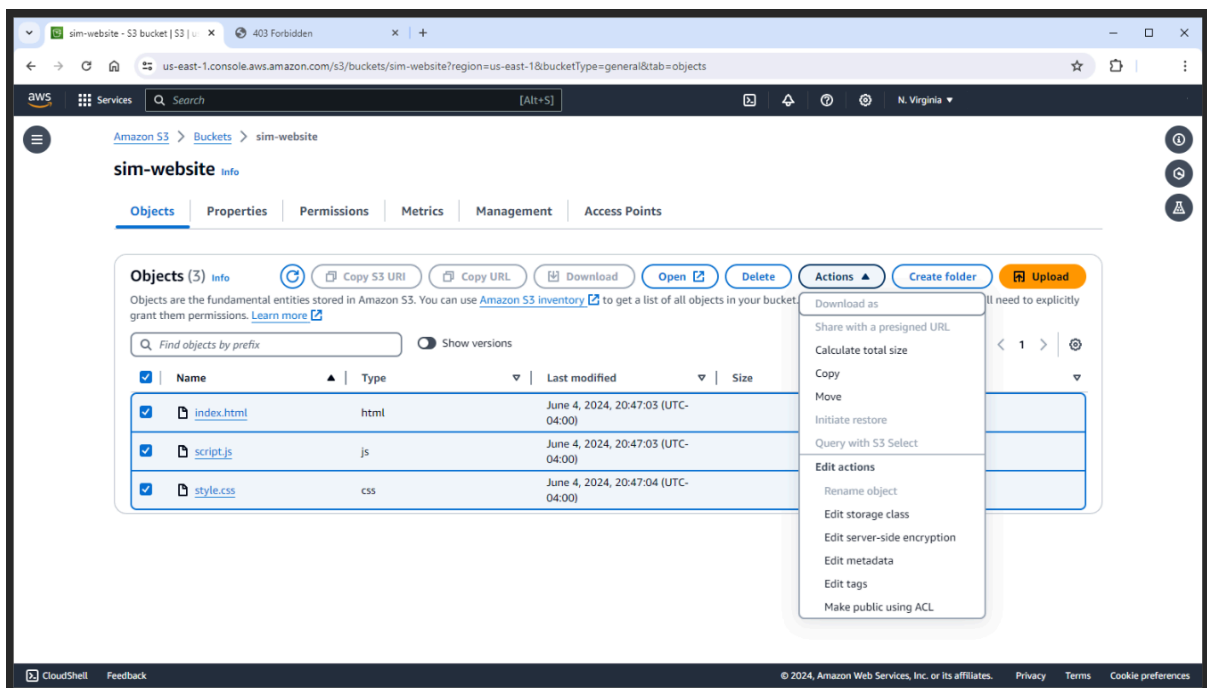
Task 4: Turning on public access to the objects

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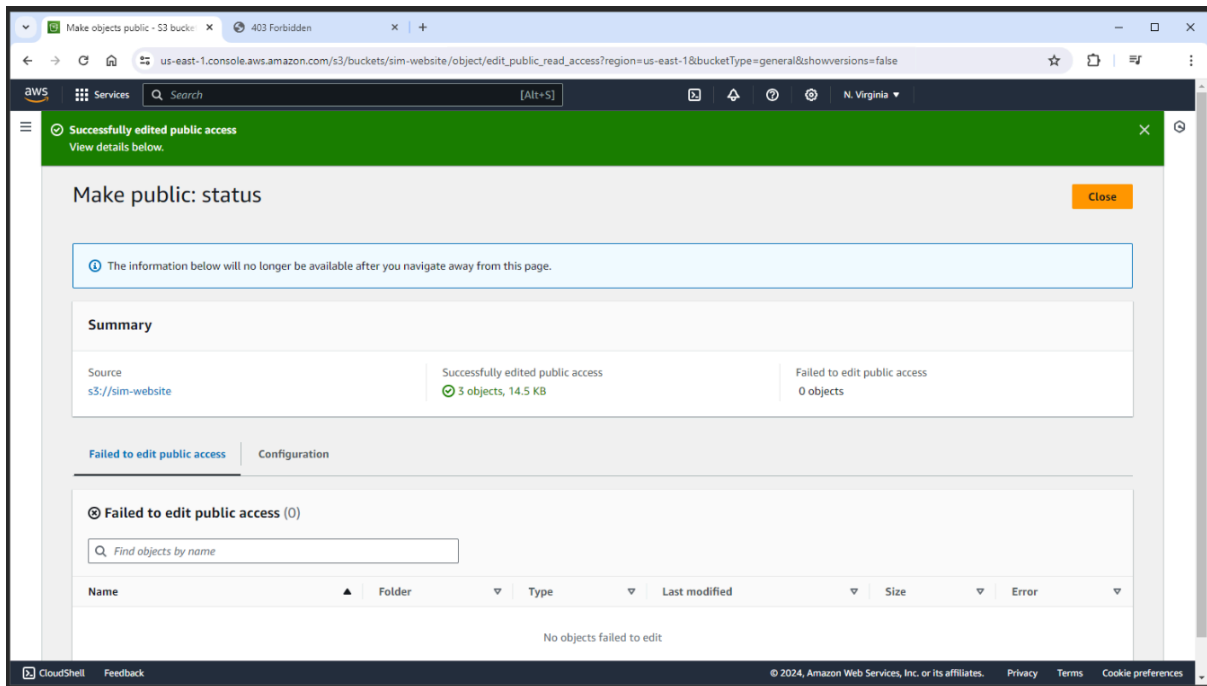
- 34.. Keep the website tab open, and return to the web browser tab with the Amazon S3 console.
35. Choose the Name checkbox to select all three objects.



36. In the Actions menu, choose Make public using ACL.
A list of the three objects is displayed.



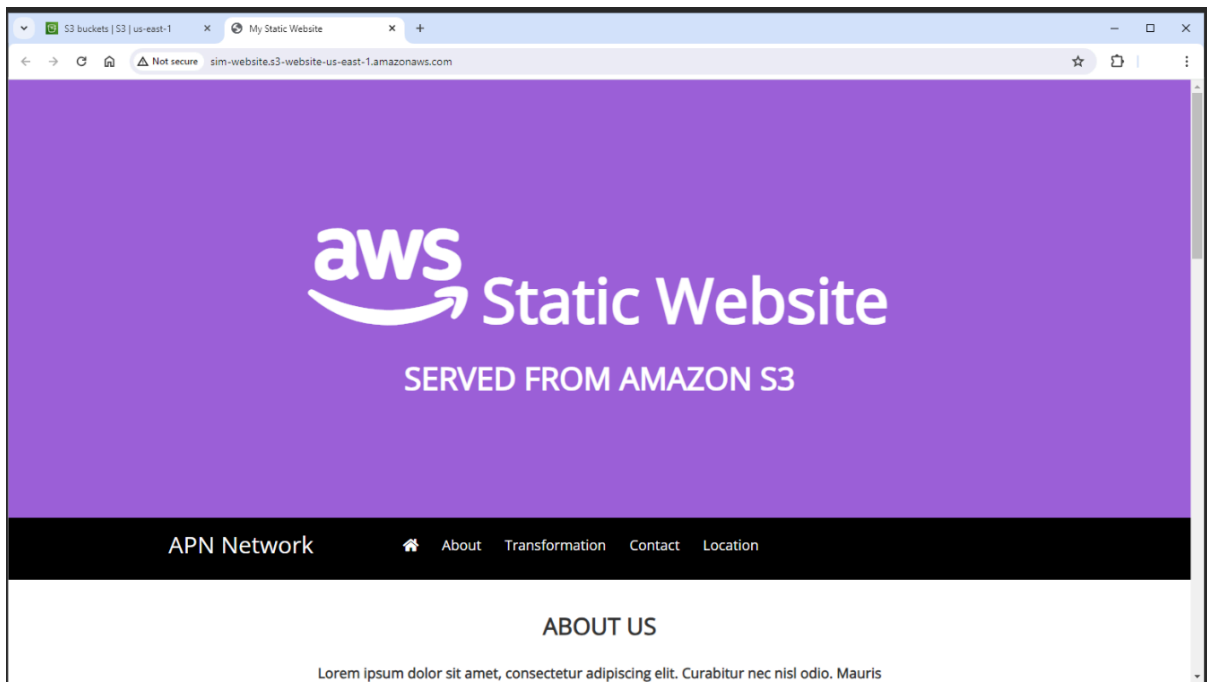
37. Choose Make public.
Your static website is now publicly accessible.
38. Choose Close.



39. Refresh the webpage.

You should now see the static website that Amazon S3 is hosting.

40. On your browser, choose the x on the My Static Website tab.



sim-website - S3 bucket | S3 | X

us-east-1.console.aws.amazon.com/s3/buckets/sim-website?region=us-east-1&bucketType=general&tab=objects

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

Feature spotlight 7

AWS Marketplace for S3

Amazon S3 > Buckets > sim-website

sim-website Info

Objects Properties Permissions Metrics Management Access Points

Objects (3) Info

Copy S3 URI Copy URL Download Open Delete Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix Show versions

☐	Name	Type	Last modified	Size	Storage class
☐	index.html	html	June 4, 2024, 20:47:03 (UTC-04:00)	9.4 KB	Standard
☐	script.js	js	June 4, 2024, 20:47:03 (UTC-04:00)	1.9 KB	Standard
☐	style.css	css	June 4, 2024, 20:47:04 (UTC-04:00)	3.1 KB	Standard

This completes Task 4: Turning on public access to the objects. Choose **Continue**.

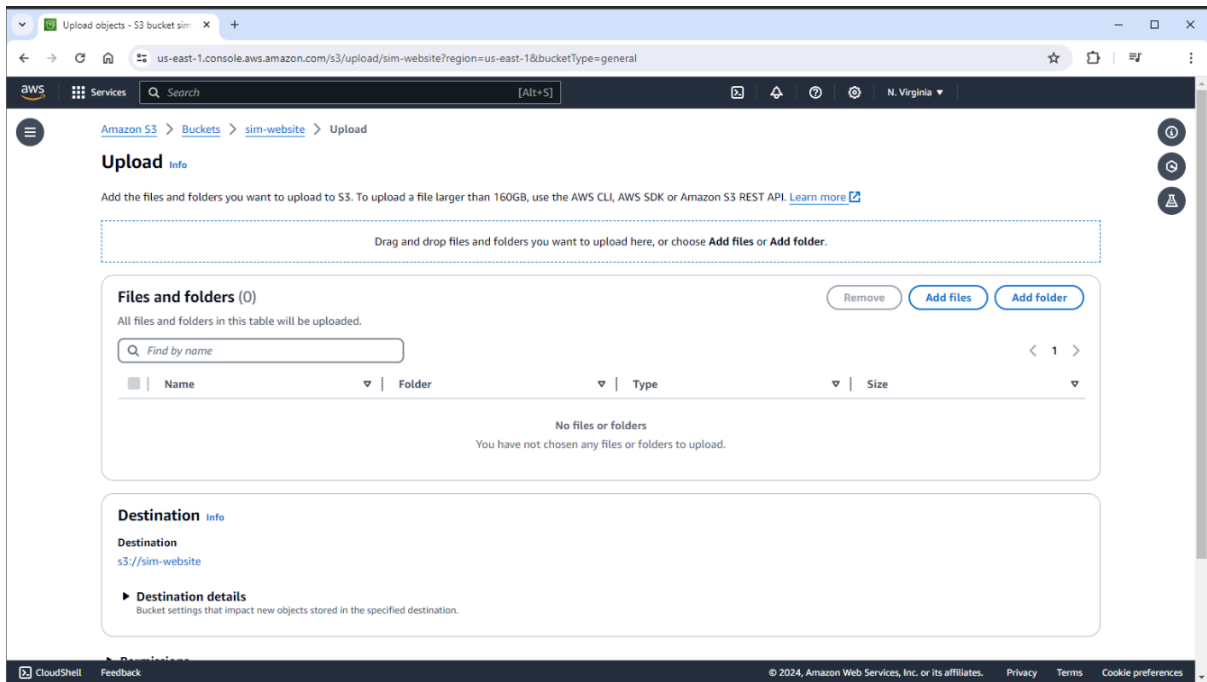
CloudShell Feedback

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Task 5: Securely sharing an object by using a presigned URL

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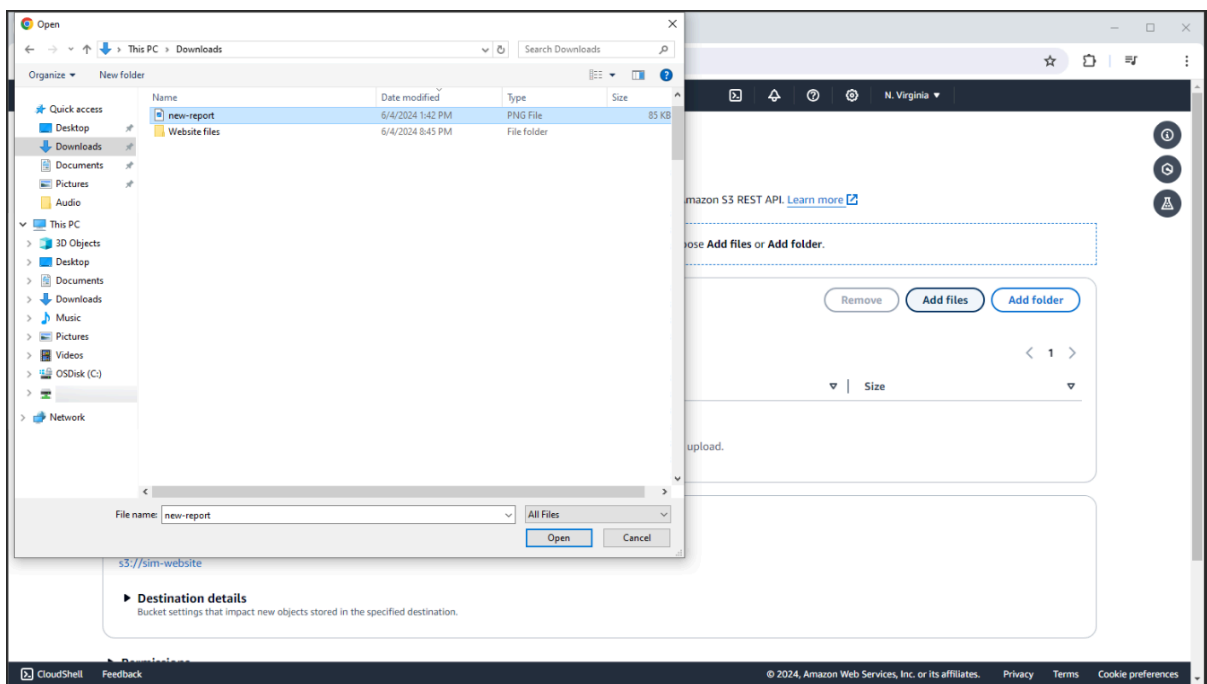
41. Choose Upload.
42. Choose Add files.

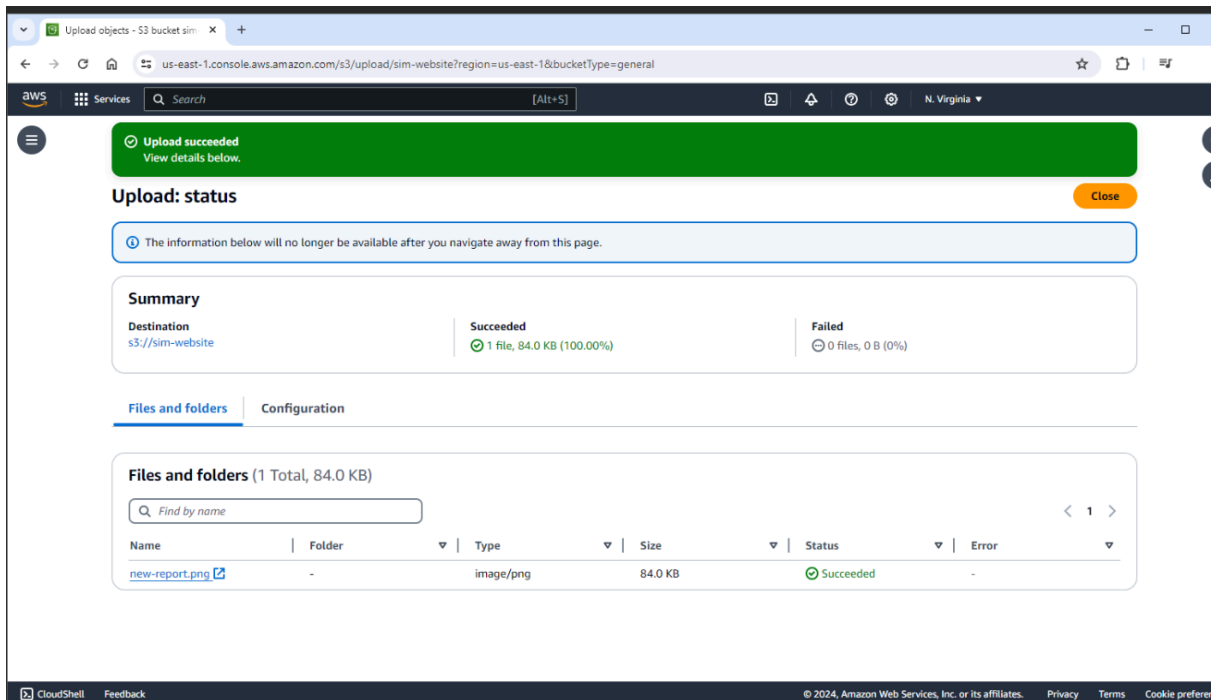


43. Choose the file new-report file and choose Open.

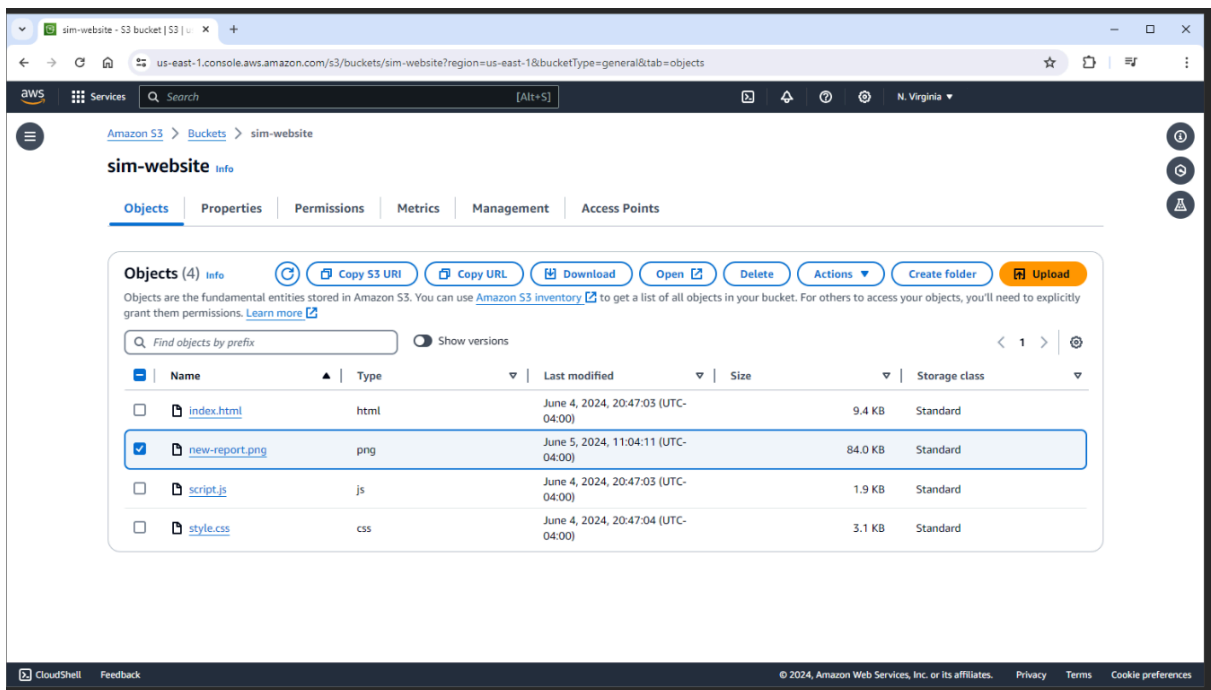
44. Choose the scroll bar to scroll down.

45. Choose Upload.





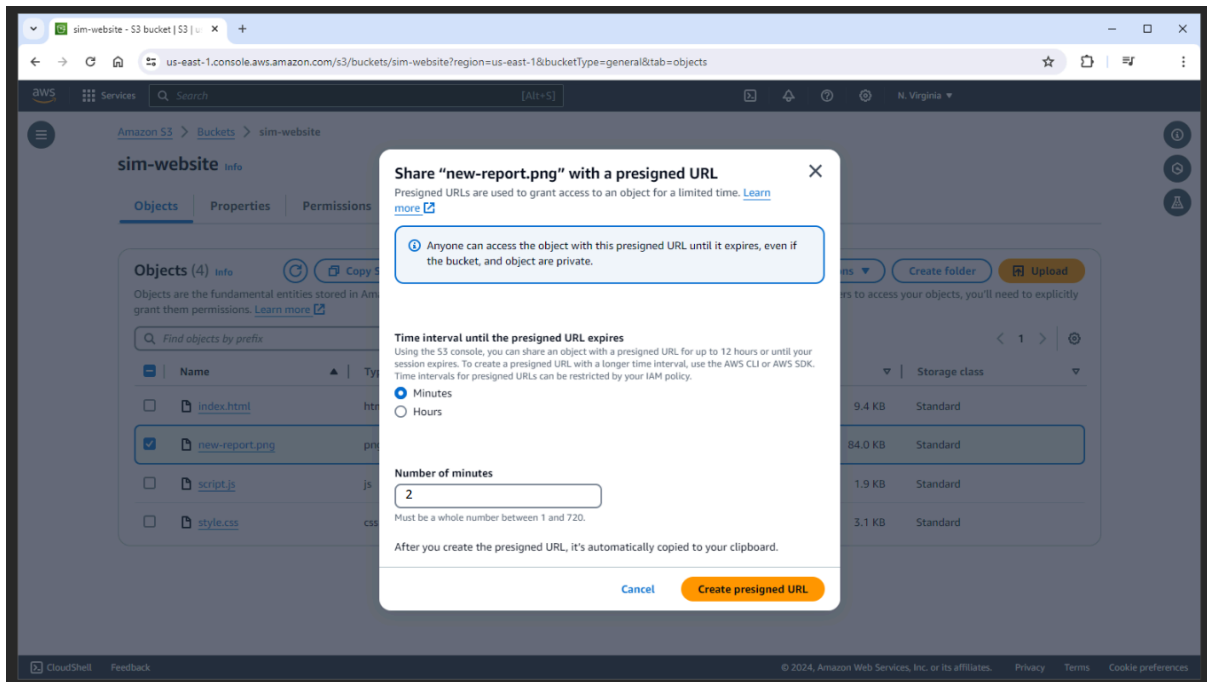
46. In the Objects tab, choose new-report.png.



47. From the Actions menu, select Share with a presigned URL.

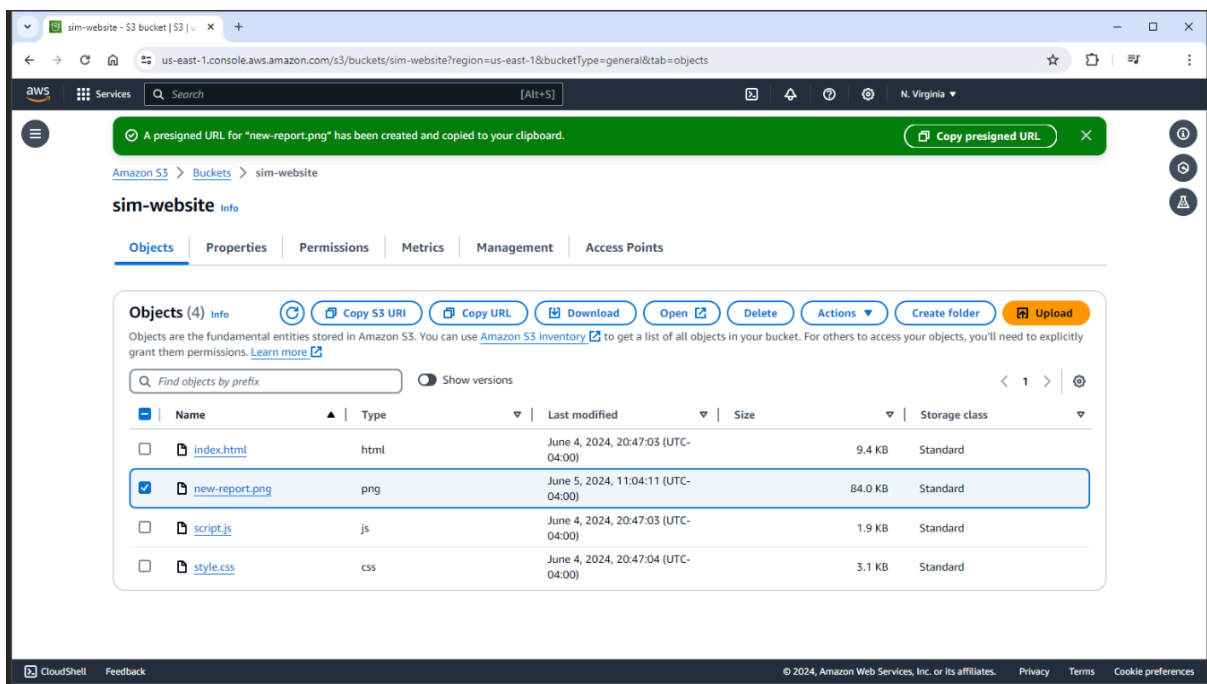
48. In the pop-up window, keep the default Minutes selected for the Time interval until the presigned URL expires.

49. For Number of minutes, enter 2.



49. Choose Create presigned URL.

50. From the banner at the top of the page, choose "Copy presigned URL."



51. Open a new browser tab.

52. Paste the URL that you copied into the address bar. Use these specific steps to paste and launch the URL:

- o Choose the browser URL search bar.
- o Press Ctrl + v on your keyboard.

sim-website - S3 bucket | S3 | X new-report.png (1045x602) X +

sim-website.s3.us-east-1.amazonaws.com/new-report.png?response-content-disposition=inline&X-Amz-Security-Token=IQoJb3JpZ2luX2VjEDcaCXVzLWVhc3QtMSJHMEUCIQCF1Lp4...

Service	Operation	UsageType	Resource	StartTime	EndTime	UsageValue
AmazonS3	HeadBucket	USW2-C3DataTransfer-Out-Bytes	lab-test-bucket-77	10/31/2020 0:00	12/31/2020 11:59	15309
AmazonS3	PutObject	USW2-C3DataTransfer-In-Bytes	admin-test-77	10/31/2020 0:00	12/31/2020 11:59	19032
AmazonS3	HeadBucket	USW2-Requests-Tier2	admin-test-77	10/31/2020 0:00	12/31/2020 11:59	128
AmazonS3	PutObjectForReplication	USW1-Request-SIA-Tier1	mybucket-98765	10/31/2020 0:00	12/31/2020 11:59	56888
AmazonS3	GetObjectForReplication	USW1-USW2-AWS-In-Bytes	mybucket-98766	10/31/2020 0:00	12/31/2020 11:59	254587
AmazonS3	GetObjectForReplication	USW2-C3DataTransfer-Out-Bytes	mybucket-98767	10/31/2020 0:00	12/31/2020 11:59	235
AmazonS3	HeadBucket	USW2-C3DataTransfer-In-Bytes	mybucket-98768	10/31/2020 0:00	12/31/2020 11:59	25589
AmazonS3	PutObject	USW2-Requests-Tier2	mybucket-98769	10/31/2020 0:00	12/31/2020 11:59	2348
AmazonS3	PutObjectForReplication	USW1-Request-SIA-Tier1	mybucket-98770	10/31/2020 0:00	12/31/2020 11:59	15309
AmazonS3	GetObjectForReplication	USW1-USW2-AWS-In-Bytes	mybucket-98771	10/31/2020 0:00	12/31/2020 11:59	19032
AmazonS3	GetObjectForReplication	USW2-C3DataTransfer-Out-Bytes	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	128
AmazonS3	HeadBucket	USW2-C3DataTransfer-In-Bytes	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	56888
AmazonS3	PutObject	USW2-Requests-Tier2	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	254587
AmazonS3	PutObjectForReplication	USW1-Request-SIA-Tier1	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	235
AmazonS3	GetObjectForReplication	USW1-USW2-AWS-In-Bytes	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	25589

sim-website - S3 bucket | S3 | X sim-website.s3.us-east-1.amazonaws.com/new-report.png?response-content-disposition=inline&X-Amz-Security-Token=IQoJb3JpZ2luX2VjEDcaCXVzLWVhc3QtMSJHMEUCIQCF1Lp4...

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Error>
  <Code>AccessDenied</Code>
  <Message>Request has expired</Message>
  <X-Amz-Expires>120</X-Amz-Expires>
  <Expires>2024-06-05T15:08:19Z</Expires>
  <ServerTime>2024-06-05T15:09:02Z</ServerTime>
  <RequestId>QID0H6NRYT81QAF3</RequestId>
  <HostId>er1UsaKe9Vnu1vq0RQ6vYdx2LWkzcc5yJ1ud3/vdcfTeXWIFKqj8YSrWic+77QPF/Y6togSd4=</HostId>
</Error>
```

sim-website - S3 bucket | S3 | X

us-east-1.console.aws.amazon.com/s3/buckets/sim-website?region=us-east-1&bucketType=general&tab=objects

A presigned URL for "new-report.png" has been created and copied to your clipboard. Copy presigned URL

Presigned URL copied

Amazon S3 > Buckets > sim-website

sim-website Info

Objects Properties Permissions Metrics Management Access Points

Objects (4) Info Copy S3 URI Copy URL Download Open Delete Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix Show versions

Name	Type	Last modified	Size	Storage class
index.html	html	June 4, 2024, 20:47:03 (UTC-04:00)	9.4 KB	Standard
☒ new-report.png	png	June 5, 2024, 11:04:11 (UTC-04:00)	84.0 KB	Standard
script.js	js	June 4, 2024, 20:47:03 (UTC-04:00)	1.9 KB	Standard
style.css	css	June 4, 2024, 20:47:04 (UTC-04:00)	3.1 KB	Standard

This completes Task 5: Securely sharing an object by using a presigned URL. Choose **Continue**.

CloudShell Feedback

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Task 6: Using a bucket policy to secure your bucket

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53. Choose the Permissions tab.
54. Choose the scroll bar to scroll down to the Bucket policy panel.
55. In the Bucket policy panel, choose Edit.

sim-website - S3 bucket | S3

us-east-1.console.aws.amazon.com/s3/buckets/sim-website?region=us-east-1&bucketType=general&tab=permissions

Services

Search

[Alt+S]

N. Virginia

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

Feature spotlight

AWS Marketplace for S3

Successfully edited bucket policy.

Amazon S3 > Buckets > sim-website

sim-website

Info

Objects | Properties | Permissions | Metrics | Management | Access Points

Permissions overview

Access finding

Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#).

[View analyzer for us-east-1](#)

Block public access (bucket settings)

Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block Public Access settings for this account are currently turned on

[Block Public Access settings for this account](#) that are enabled apply even if they are disabled for this bucket.

Block all public access

Off

[Individual Block Public Access settings for this bucket](#)

CloudShell

Feedback

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Delete objects - S3 bucket sim-website

us-east-1.console.aws.amazon.com/s3/buckets/sim-website/object/delete?region=us-east-1&bucketType=general&showversions=false

Services

Search

[Alt+S]

N. Virginia

Failed to delete objects

For more information, see the Error column in the Failed to delete table below.

[Diagnose with Amazon Q](#)

Close

Delete objects: status

The information below will no longer be available after you navigate away from this page.

Summary

Source

s3://sim-website

Successfully deleted

0 objects

Failed to delete

1 object, 9.4 KB

Failed to delete | Configuration

Failed to delete (1 object, 9.4 KB)

Name	Folder	Type	Last modified	Size	Error
index.html	-	html	June 4, 2024, 20:47:03 (UTC-04:00)	9.4 KB	Access denied

CloudShell

Feedback

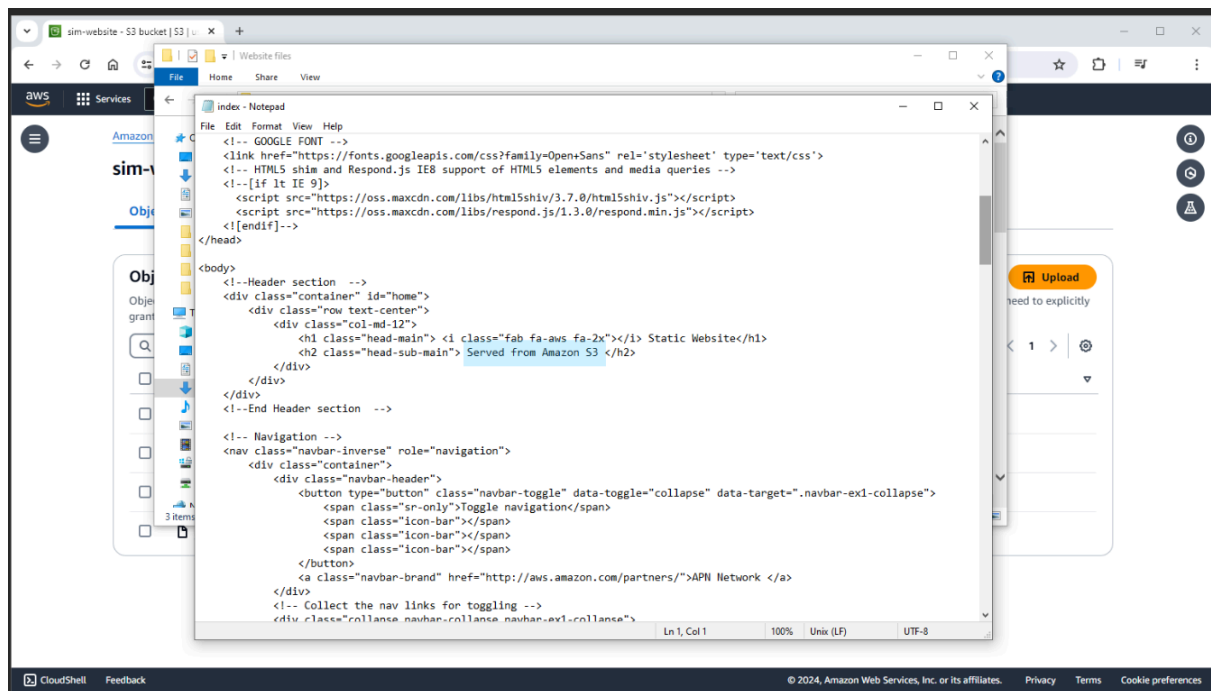
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Task 7: Updating the website

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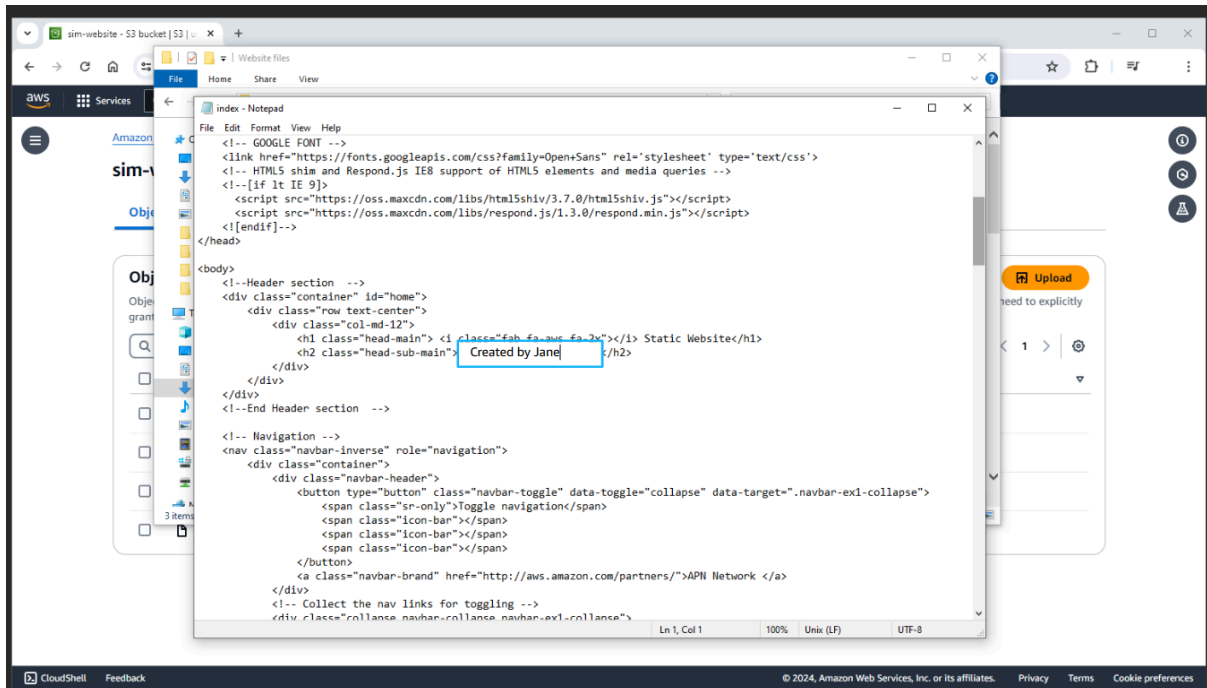
56. On your computer, load the `index.html` file into a text editor (in this simulation, you use Notepad). Follow these specific steps:

1. Open the context (right-click) menu for the `index.html` file.
2. Choose Open with.
3. Choose Notepad.



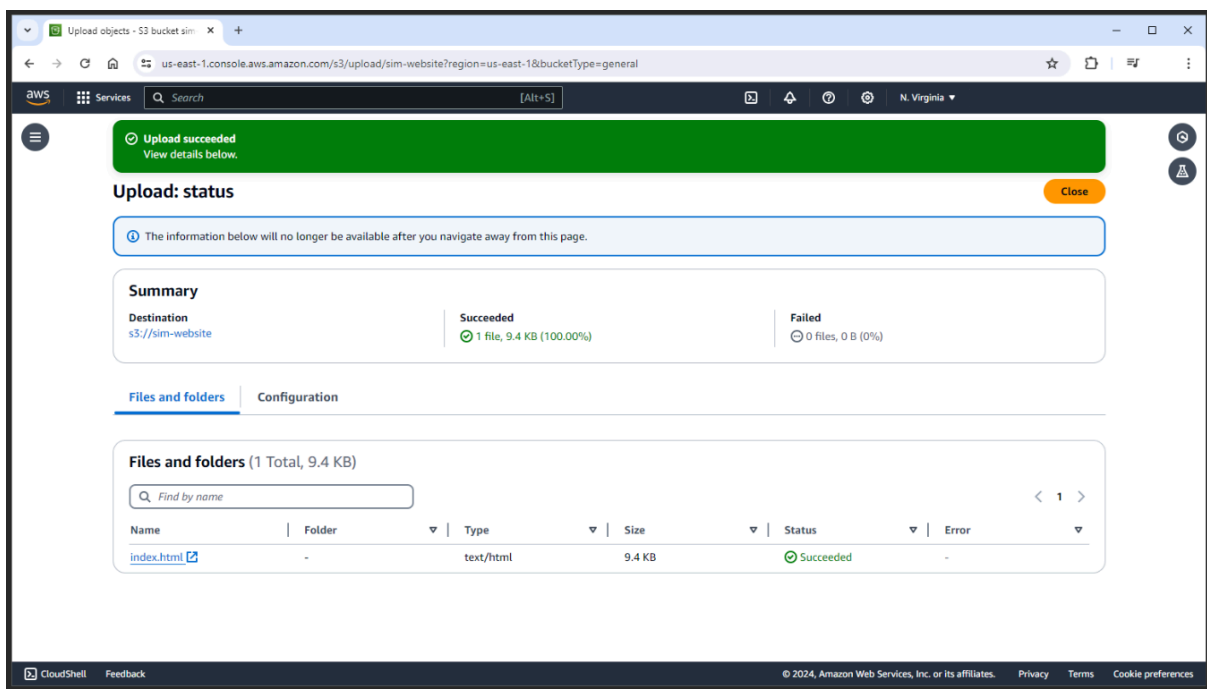
57. Find the text `Served from Amazon S3`, and replace it with `Created by Jane`. Follow these specific steps:

1. Choose the text Served from Amazon S3.
2. Enter Created by Jane.

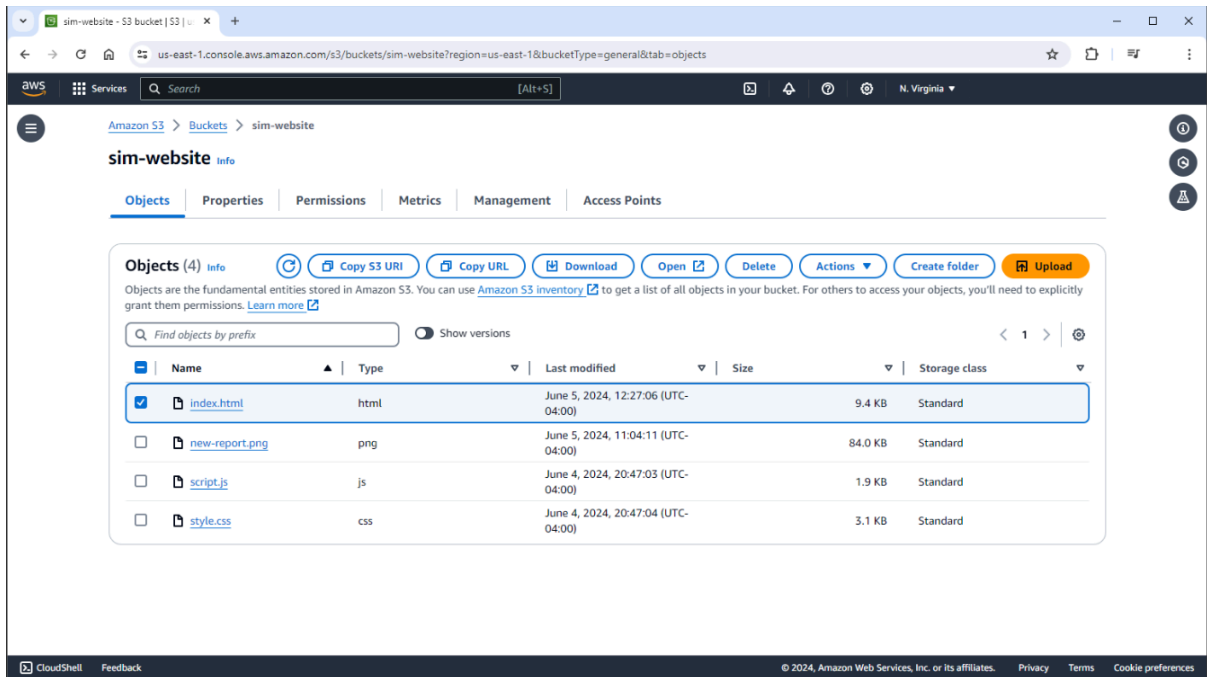


58. Save the file. Follow these specific steps:

1. Choose File from the Notepad menu.
2. Choose Save.



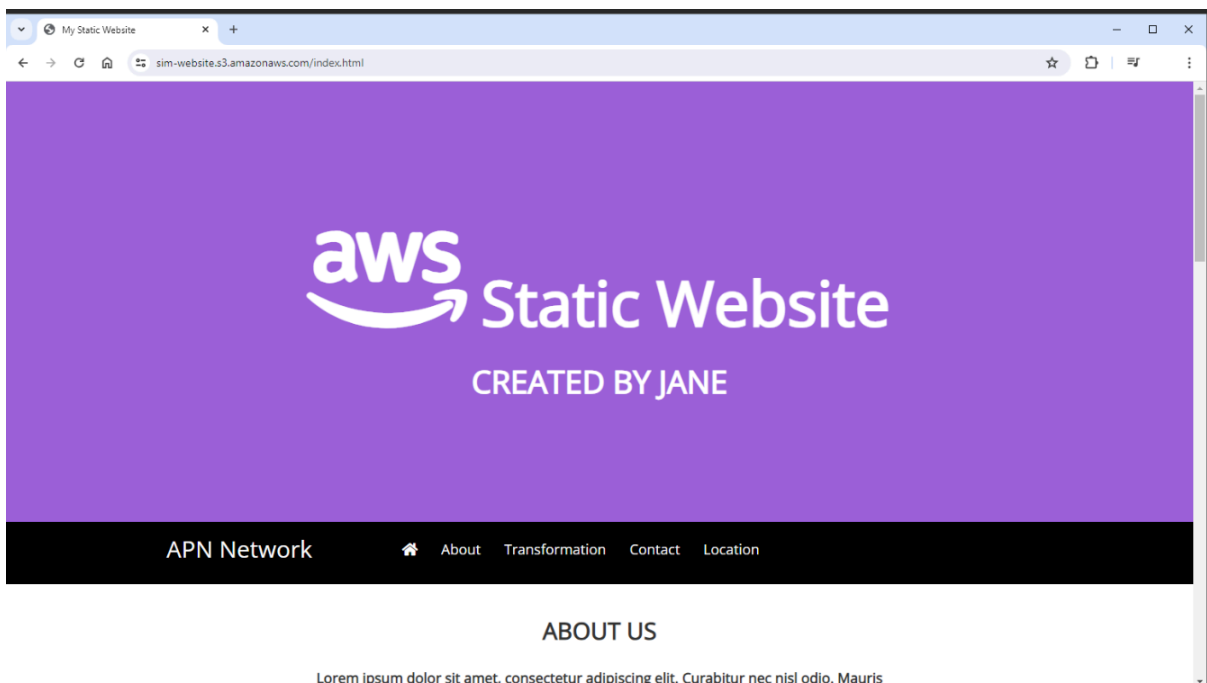
59. Choose the index.html file name (choose the link, not the checkbox).
60. Choose the Object URL link.

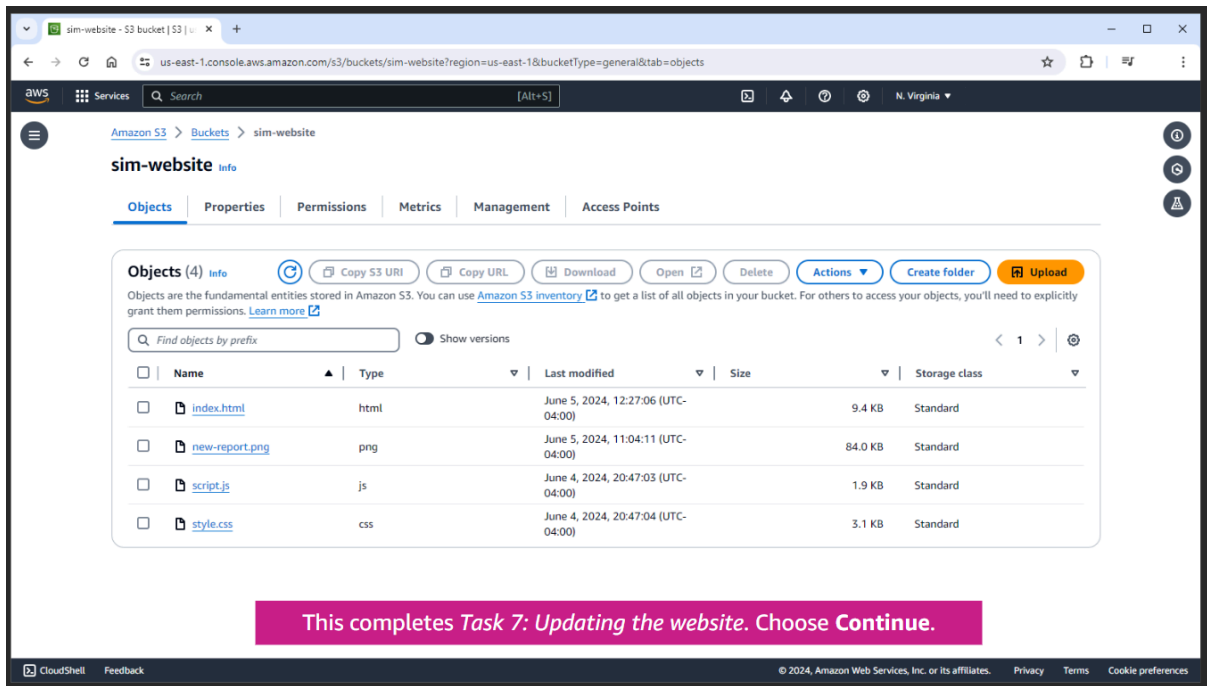


Created by Jane should now be on the page in place of Served from Amazon S3. Your static website is now accessible on the internet. Because it is hosted on Amazon S3, the website has high availability and can serve high volumes of traffic without using any servers.

61. Choose the Back arrow on your browser to return to the Amazon S3 console.

62. Choose the sim-website link from the navigation at the top of the page.





63. Choose Show versions to see which files have multiple versions.

64. Choose the scroll bar to scroll down.

65. Review the list of objects in the bucket.

o Notice that each file has a Version ID. Amazon S3 automatically generates these IDs when versioning is turned on.

o You should also find two versions of the index.html file because you uploaded a new version of the file. The current version is the file that you uploaded when you updated your website.

